INDIAN INSTITUTE OF TECHNOLOGY, HYDERABAD

GREENKO SCHOOL OF SUSTAINABILITY

भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS OF THE COURSE

SE5713: ANALYSIS METHODS IN SUSTAINABLE ENGINEERING

Optimisation of Multi-source Energy Allocation for Mumbai's Urban Energy Transition (2025-2050)

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1. CONTEXT

This study develops a structured optimization model to support long-term energy and resource planning across the horizon years **2030, 2040, and 2050** for the city of **Mumbai,** aligning with the **Mumbai Climate Action Plan 2022.** Rapid urban growth, rising electricity demand, and the increasing need for low-carbon transitions require systematic tools that can identify the most **cost-effective and sustainable combinations of energy sources.**

The project uses a **linear optimization** model built and executed in **LINGO** to determine the optimal energy mix and resource allocation required to satisfy projected demands under the constraints. **Costs, emission factors, renewable energy targets, and technology capacities change with time**, so assessing each decade separately helps reveal how system priorities shift.

By comparing outcomes for 2030, 2040, and 2050, the study highlights **trade-offs** between **economic and environmental goals**, identifies conflicting constraints, and examines how future policies shape feasible solutions. The final outputs **provide planners and policy makers** with a clear understanding of **optimal strategies and long-term implications** for clean and green energy optimization strategies in Mumbai.

2. DELIVERABLES

2.1. OVERVIEW

Our objective is to produce a structured technical report and supporting files (model code and data). The deliverables will include:

1. A clear **forecasting procedure to estimate annual electricity demand** in 2025 and 2050 (with intermediate yearly projections 2030, 2040 and 2050).
2. A fully formulated **multi-objective optimisation model** (mathematical formulation with objective functions and constraints).
3. Implementation-ready formulation for a **solver (LP) and LINGO code.**
4. Solution for three scenarios (2030, 2040, 2050) with results tables and interpretation.

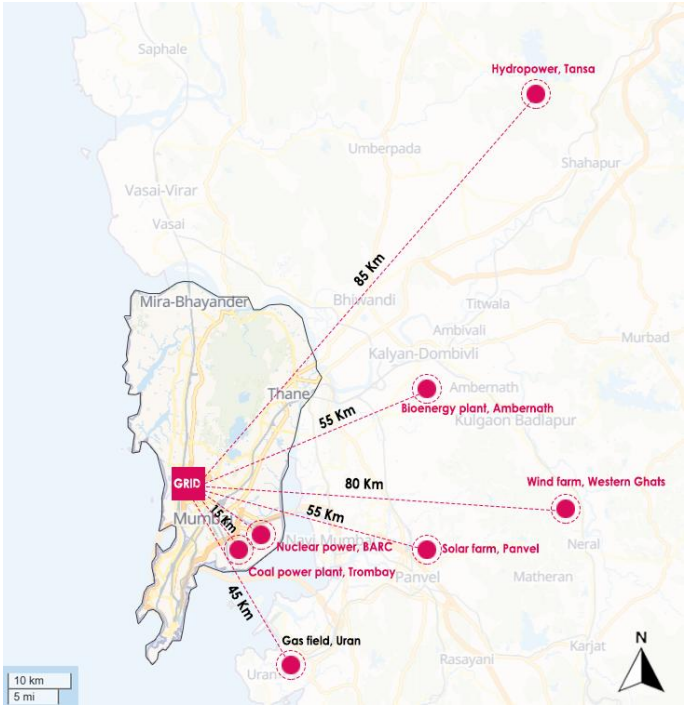

Figure 1: Mumbai Map with marked energy sourcing locations

2.2. PROBLEM STATEMENT

Mumbai's electricity system is standing at a crossroads. Every year, the city's demand rises sharply, driven by growing commercial districts, expanding transport networks, and an increasing dependence on cooling and digital infrastructure. Yet the city **still relies heavily on coal-based power transported from distant thermal plants, contributing to severe CO_2, NO_x, and SO_x emissions.** This not only worsens Mumbai's air quality but also directly contradicts the long-term decarbonization targets laid out in the Mumbai Climate Action Plan (MCAP 2022).

At the same time, climate variability is already reshaping how Mumbai receives energy. **Summers are becoming hotter, increasing electricity demand; monsoons are becoming more unpredictable, affecting renewable generation from solar and wind.** Rising fuel prices and transport distances make thermal power economically vulnerable, while intermittent renewables need careful planning to ensure reliability.

Policy mandates such as minimum renewable share, caps on coal usage, and the strategic role of nuclear and hydropower add another layer of complexity. Mumbai must meet

rising demand, lower emissions, reduce dependence on volatile fossil fuel supply chains, and maintain affordability for consumers all at once.

This creates a classic multi-criteria challenge: ***How can Mumbai simultaneously minimize cost and reduce emissions while satisfying strict policy constraints and seasonal renewable variability?***

To address this, a mathematical optimization model becomes essential. By formulating the problem as a linear program with weighted objectives, we can determine the most cost-effective and low-carbon combination of coal, gas, nuclear, hydro, solar, and wind, distributed seasonally based on real-world generation patterns. **This approach allows Mumbai to transition from a fossil-heavy supply mix toward a balanced, resilient, and sustainable energy future backed by data rather than intuition.**

We are using a combination of real data (from NITI Aayog) and additional data was taken from the literature review.

1. City's current population (2025): 24.25 million.
2. Projected population (2050): 30.5 million.
3. Annual electricity demand (2025 baseline): 4776 MW.
4. Projected demand increases to 2050: +300% (i.e., 14,321.25 MW) due to population increase and temperature rise (mean +2.2°C).
5. Energy sources considered:
    a) **Coal:** Trombay coal power plant, Mumbai Suburban district.
    b) **Solar PV**: Panvel Solar PV farm, Raigad district.
    c) **Onshore Wind:** Western Ghats, Raigad district.
    d) **Hydro-electric power plant**: Tansa river reservoir, Thane district.
    e) **Bioenergy** (residues and municipal solid waste): Ambernath, Thane district.
    f) **Nuclear**: BARC, Mumbai Suburban district
    g) **Natural gas**: Uran gas field, Raigad district.
6. Time horizon: Decadal planning with 3 target years (2030, 2040 and 2050).

2.3. ASSUMPTIONS

1. Constant rate of cost inflation: The cost of electricity generation and transmission is assumed to be 7% till 2050.
2. Policies: The policy constraints mentioned in section 4.3 align with MCAP 2022.

3. FORECASTING THE ENERGY DEMAND

1. Using a multiple linear regression model, we will forecast the **annual electricity demand** $\boldsymbol{E_t}$ as a function of **population** $\boldsymbol{P_t}$, **mean annual temperature** $\boldsymbol{T_t}$ and the number of **electric vehicles** $\boldsymbol{EV_n}$.

$$\boldsymbol{E_t = \alpha + \beta_1 P_t + \beta_2 T_t + \beta_3 EV_n}$$

2. Calibrate our model using historical city-level data (1950–2024). If city-level historical data are unavailable, then we will adhere to synthetically generated data.
3. We shall provide forecasts and a 95% confidence level for E_{2030} , E_{2040} and E_{2050}.

Results of the regression are discussed in chapter 8.

4. DEVELOPING THE OPTIMIZATION MODEL

Formulating a multi-objective linear program (LP) to determine the generation by technology MWh for years **2030, 2040 and 2050.** Also, we will provide appropriate proportions of energy capacity additions.

4.1. DECISION VARIABLES

1. $\boldsymbol{X_{coal}}$**:** Energy (MWh) generated by coal power plants.
2. $\boldsymbol{X_{solar}}$: Energy (MWh) generated by solar power.
3. $\boldsymbol{X_{wind}}$: Energy (MWh) generated by wind power.
4. $\boldsymbol{X_{hydro}}$**:** Energy (MWh) generated by hydroelectric power plants.
5. $\boldsymbol{X_{biomass}}$**:** Energy (MWh) generated by biomass.
6. $\boldsymbol{X_{nuclear}}$: Energy (MWh) generated by nuclear power plants.
7. $\boldsymbol{X_{ng}}$: Energy (MWh) generated by Natural gas power plants.

4.2. OBJECTIVE FUNCTION

4.2.1. PARAMETERS

1. $\boldsymbol{C_i}$: Levelized cost of electricity (₹/MWh)
2. $\boldsymbol{f_{CO_2}}$: Emission factor (kg CO_2 /MWh)
3. $\boldsymbol{f_{NO_2}}$: Emission factor (kg NO_2/ MWh)
4. $\boldsymbol{f_{SO_2}}$: Emission factor (kg SO_2/ MWh)
5. $\boldsymbol{C_{ti}}$: Cost of electricity transmission to the grid (₹/MWh /km)

4.2.2. ASSUMPTIONS FOR WEIGHTAGE

1. **60% weightage** has been allotted to environmental impact-related objective functions.
2. **Rest 40% weightage** has been given to economically related objective functions.

$$\mathbf{min}\ \boldsymbol{cost}: \boldsymbol{Z_1} = \sum_i \boldsymbol{C_i X_i}$$

$$\mathbf{min}\ \boldsymbol{CO_2\ emissions}: \boldsymbol{Z_2} = \sum_i \boldsymbol{f_{CO_2} X_i}$$

$$\mathbf{min}\ \boldsymbol{Conveyance\ cost}: \boldsymbol{Z_3} = \sum_i \boldsymbol{C_{ti} X_i}$$

$$\mathbf{min}\ \boldsymbol{NO_x\ emissions}: \boldsymbol{Z_4} = \sum_i \boldsymbol{f_{NO_2} X_i}$$

$$\mathbf{min}\ \boldsymbol{SO_x\ emissions}: \boldsymbol{Z_5} = \sum_i \boldsymbol{f_{SO_2} X_i}$$

4.2.3. SINGLE OBJECTIVE FUNCTION

$$\mathbf{Minimize}\ \boldsymbol{Z} = \mathbf{0.6}(\boldsymbol{Z_2} + \boldsymbol{Z_4} + \boldsymbol{Z_5}) + \mathbf{0.4}(\boldsymbol{Z_1} + \boldsymbol{Z_3})$$

4.3. CONSTRAINTS

1. Meeting minimum energy demand for the years **2030, 2040, and 2050**

$$\sum_i X_i \geq E_t$$

2. COP26 renewable target

India is committed to source **50%** of its energy from renewables by 2030. Constraints align with the **Mumbai Climate Action Plan 2022**.

3. Policy constraints

- **Constraints for 2030**
  - **Renewable minimum share:** $X_{\text{solar}} + X_{\text{wind}} + X_{\text{hydro}} + X_{\text{biomass}} \geq 0.5E_t$ (At least 50% of the demand in 2030 must be met by renewables)
  - **Solar:** $X_{\text{solar}} \geq 0.10E_t$ (At least 10% of total demand to be satisfied by solar energy in 2030)
  - **Wind:** $X_{\text{wind}} \geq 0.05E_t$ (At least 5% of total demand to be satisfied by wind energy in 2030)
  - **Hydro:** $X_{\text{hydro}} \geq 0.15E_t$ (At least 15% of total demand to be satisfied by hydro energy in 2030)
  - **Coal cap:** $X_{\text{coal}} \leq 0.35E_t$ (Coal power plants capped at 35% of total energy demand in 2030)
  - **Nuclear**: $X_{\text{nuclear}} \geq 0.15E_t$ (Nuclear energy to satisfy total energy demand by not less than 15%)
- **Constraints for 2040**
  - **Renewable minimum share:** $X_{\text{solar}} + X_{\text{wind}} + X_{\text{hydro}} + X_{\text{biomass}} \geq 0.7E_t$ (At least 70% of the demand in 2040 must be met by renewables)
  - **Solar:** $X_{\text{solar}} \geq 0.25E_t$ (At least 25% of total demand to be satisfied by solar energy in 2040)
  - **Wind:** $X_{\text{wind}} \geq 0.12E_t$ (At least 12% of total demand to be satisfied by wind energy in 2040)
  - **Hydro:** $X_{\text{hydro}} \geq 0.20E_t$ (At least 20% of total demand to be satisfied by hydro energy in 2040)

- **Coal cap:** $X_{\text{coal}} \leq 0.20E_t$ (Coal power plants capped at 20% of total energy demand in 2040)
- **Nuclear**: $X_{\text{nuclear}} \geq 0.15E_t$ (Nuclear energy to satisfy total energy demand by not less than 15%)

- **Constraints for 2050**
  - **Renewable minimum share**: $X_{\text{solar}} + X_{\text{wind}} + X_{\text{hydro}} + X_{\text{biomass}} \geq 0.9E_t$ (At least 90% of the demand in 2050 must be met by renewables)
  - **Solar**: $X_{\text{solar}} \geq 0.4E_t$ (At least 40% of total demand to be satisfied by solar energy in 2050)
  - **Wind**: $X_{\text{wind}} \geq 0.15E_t$ (At least 10% of total demand to be satisfied by wind energy in 2050)
  - **Hydro**: $X_{\text{hydro}} \geq 0.25E_t$ (At least 25% of total demand to be satisfied by hydro energy in 2050)
  - **Coal cap**: $X_{\text{coal}} \leq 0.05E_t$ (Coal power plants to be capped at 5% of the total energy demand in 2050)
  - **Nuclear**: $X_{\text{nuclear}} \geq 0.15E_t$ (Nuclear energy to satisfy total energy demand by not less than 15%)

4. Resource availability

- Biomass capacity**:** $\sum_i X_{\text{biomass}} \leq$ **Available biomass capacity** $(0.155$ **GW**$)$
- Solar, Quarter 1: $\sum_i X_{\text{solar,1}} \leq 13.96272229$ **GW**
- Solar, Quarter 2: $\sum_i X_{\text{solar,2}} \leq 19.60542384$ **GW**
- Solar, Quarter 3: $\sum_i X_{\text{solar,3}} \leq 16.1586875$ **GW**
- Solar, Quarter 4: $\sum_i X_{\text{solar,4}} \leq 14.42845665$ **GW**
- Wind, Quarter 1: $\sum_i X_{\text{wind,1}} \leq 5.29793$ **GW**
- Wind, Quarter 2: $\sum_i X_{\text{wind,2}} \leq 0.759303$ **GW**
- Wind, Quarter 3**:** $\sum_i X_{\text{wind,3}} \leq 10.87625$ **GW**
- Wind, Quarter 4: $\sum_i X_{\text{wind,4}} \leq 2.216771$ **GW**

To incorporate seasonality, **fixed fractional distributions were assigned to total annual solar and wind generation.** For solar, the model distributes **35% in summer, 15% in monsoon, 30% in autumn, and 20% in winter**, reflecting higher irradiation in summer and reduced output during monsoon. For wind, the fractions were set as **15% in summer, 45% in monsoon, 25% in autumn, and 15% in winter**, capturing the strong south-westerly monsoon winds and comparatively lower wind speeds in other seasons.

5. EMISSIONS AND COST DATA

Energy Source	Installed Capacity 2025 (GW)	Generation 2025 (TWh)	CO_2 (kg/MWh)	NO_x (kg/MWh)	SO_x (kg/MWh)	CAPEX (₹ crore/ MW)	O&M (% of CAPEX/year)	Conveyance Cost (₹/MWh/km)	LCOE (₹/MWh)
Coal	3	18.396	900	2.4	2.8	6	4	0.75	4
Natural Gas	1	4.38	450	1.2	0.3	5	3.5	0.7	3.5
Solar PV	2	3.504	45	0.05	0.02	3.5	1.5	0.5	2.2
Wind	0.5	0.788	12	0.03	0.01	5.5	2	0.55	2.5
Hydro	1.5	5.913	24	0.04	0.03	9	2.5	0.65	3
Nuclear Energy	0.5	3.723	15	0.02	0.02	15	3	0.8	3.2
Biomass	0.2	0.876	230	0.25	0.35	7	5	0.6	3.8

Table 1: Emissions and cost data

6. TRAINING DATASET (DEMAND DATASET)

Year	Population (Millions)	Average Temperature (°C)	Number of EVs	Standardized population	Standardized Temperature	Standardized EVs	Forecasted Demand
2000	18000000	26.89	2200	2520	-1.68168	-1.84559	-0.67837
2001	18250000	27.11	2479	2597	-1.61441	-1.19868	-0.67712
2002	18500000	26.97	2796	2675	-1.54715	-1.61035	-0.6757
2003	18750000	27.16	3153	2753	-1.47988	-1.05166	-0.6741
2004	19000000	26.97	3555	2832	-1.41261	-1.61035	-0.67229
2005	19250000	27.15	4008	2911	-1.34535	-1.08106	-0.67026
2006	19500000	27.23	4519	2991	-1.27808	-0.84582	-0.66798
2007	19750000	27.07	5095	3071	-1.21081	-1.3163	-0.66539
2008	20000000	27.13	5744	3152	-1.14354	-1.13987	-0.66248
2009	20250000	27.08	6477	3233	-1.07628	-1.28689	-0.6592
2010	20500000	27.18	7304	3316	-1.00901	-0.99285	-0.65549
2011	20750000	27.53	8235	3399	-0.94174	0.036324	-0.65132
2012	21000000	27.27	9284	3484	-0.87447	-0.7282	-0.64662
2013	21250000	27.08	10468	3570	-0.80721	-1.28689	-0.64131
2014	21500000	27.6	11803	3657	-0.73994	0.242158	-0.63533
2015	21750000	26.99	13308	3745	-0.67267	-1.55154	-0.62859
2016	22000000	27.39	15004	3836	-0.60541	-0.37534	-0.62098
2017	22250000	27.49	16918	3928	-0.53814	-0.0813	-0.61241
2018	22500000	27.27	19076	4023	-0.47087	-0.7282	-0.60273
2019	22750000	27.46	21507	4120	-0.4036	-0.16951	-0.59184
2020	23000000	27.42	24251	4220	-0.33634	-0.28713	-0.57954
2021	23250000	27.55	27342	4323	-0.26907	0.095133	-0.56569
2022	23500000	27.37	30829	4430	-0.2018	-0.43415	-0.55006
2023	23750000	27.55	34758	4540	-0.13453	0.095133	-0.53245
2024	24000000	27.42	39191	4656	-0.06727	-0.28713	-0.51258
2025	24250000	27.6	44187	4776	0	0.242158	-0.49019

Table 2: Training dataset with standardized parameters

7. CORRELATION MATRIX FOR POPULATION DATASET

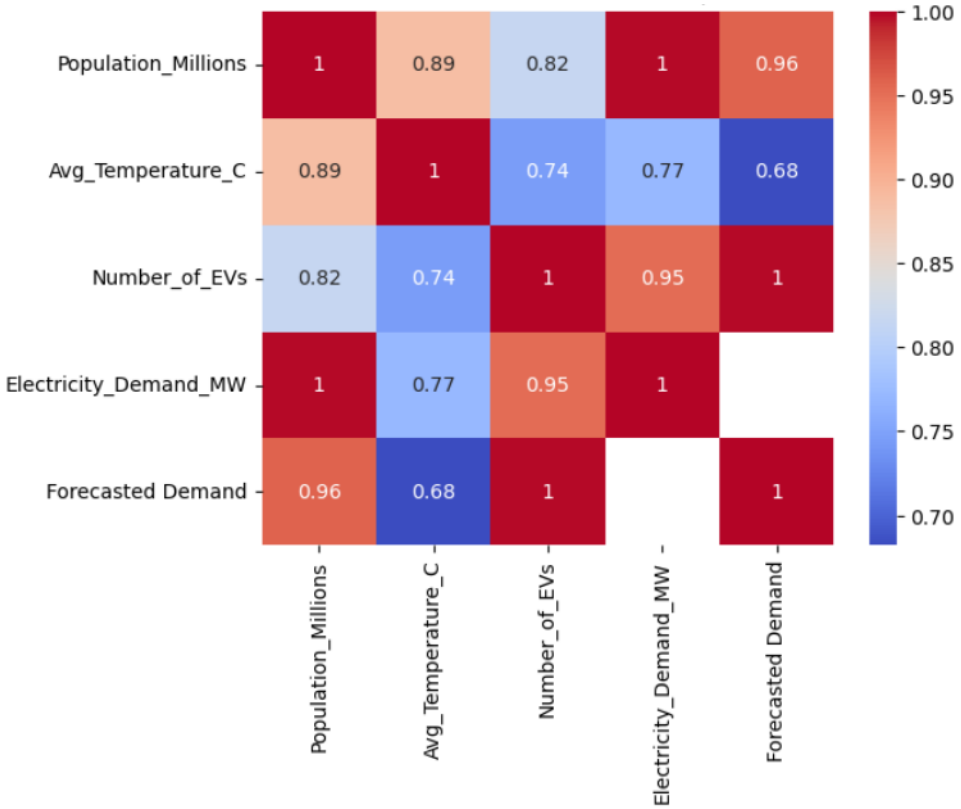

Figure 2: Correlation matrix

8. REGRESSION RESULTS

8.1. REGRESSION RESULTS

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     Electricity_Demand_MW   R-squared:                       1.000
Model:                               OLS   Adj. R-squared:                  1.000
Method:                    Least Squares   F-statistic:                 5.380e+07
Date:                   Tue, 18 Nov 2025   Prob (F-statistic):           1.17e-75
Time:                           23:14:05   Log-Likelihood:                -0.40123
No. Observations:                     26   AIC:                             8.802
Df Residuals:                         22   BIC:                             13.83
Df Model:                              3
Covariance Type:               nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const           5770.9052      1.452   3974.498      0.000    5767.894    5773.916
Std population  1114.7914      0.317   3521.680      0.000    1114.135    1115.448
std temp          -0.0836      0.131     -0.636      0.531      -0.356       0.189
std Evs         2029.3258      2.682    756.510      0.000    2023.763    2034.889
==============================================================================
Omnibus:                        2.050   Durbin-Watson:                   2.650
Prob(Omnibus):                  0.359   Jarque-Bera (JB):                1.807
Skew:                          -0.563   Prob(JB):                        0.405
Kurtosis:                       2.368   Cond. No.                         100.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

$$E_t = \alpha + \beta_1 P_t + \beta_2 T_t + \beta_3 EV_n$$

Where $\alpha = 5770.9052$ $\beta_1 = 1114.7914$ $\beta_2 = -0.0836$ $\beta_3 = 2029.3258$

So, the final equation becomes:

$$E_t = 5770.9052 + 1114.7914 * P_t + (-0.0836 * T_t) + 2029.3258 * EV_n$$

8.2. FITTING THE TEST DATA

Year	Population (Millions)	Average Temperature (°C)	Number of EVs	Standardized population	Standardized Temperature	Standardized EVs	Forecasted Demand
2026	24500000	27.69	49821	0.067267	0.506801	-0.46494	4902.341
2027	24750000	27.61	56173	0.134535	0.271562	-0.43647	5035.116
2028	25000000	27.73	63336	0.201802	0.624421	-0.40437	5175.225
2029	25250000	27.59	71410	0.269069	0.212753	-0.36818	5323.683
2030	25500000	27.68	80516	0.336336	0.477396	-0.32737	5481.468
2031	25750000	27.55	90781	0.403604	0.095133	-0.28136	5649.852
2032	26000000	27.39	102355	0.470871	-0.37534	-0.22949	5830.149
2033	26250000	27.77	115405	0.538138	0.74204	-0.171	6023.739
2034	26500000	27.67	130119	0.605406	0.447992	-0.10506	6232.574
2035	26750000	27.88	146709	0.672673	1.065493	-0.0307	6458.401
2036	27000000	27.6	165414	0.73994	0.242158	0.053128	6703.578
2037	27250000	27.56	186503	0.807207	0.124538	0.147647	6970.385
2038	27500000	27.57	210283	0.874475	0.153943	0.254224	7261.652
2039	27750000	28.03	237094	0.941742	1.506566	0.374388	7580.38
2040	28000000	27.89	267322	1.009009	1.094898	0.509865	7930.33
2041	28250000	27.73	301404	1.076276	0.624421	0.662617	8315.342
2042	28500000	28.07	339834	1.143544	1.624186	0.834852	8739.769
2043	28750000	27.8	383161	1.210811	0.830255	1.029036	9208.886
2044	29000000	27.47	432012	1.278078	-0.14011	1.247978	9728.261
2045	29250000	27.78	487093	1.345346	0.771445	1.494844	10304.15
2046	29500000	27.89	549197	1.412613	1.094898	1.773183	10943.95
2047	29750000	28.13	619216	1.47988	1.800615	2.086999	11655.71
2048	30000000	28.31	698166	1.547147	2.329902	2.440836	12448.71
2049	30250000	28.04	787180	1.614415	1.535971	2.839784	13333.36
2050	30500000	28.04	887542	1.681682	1.535971	3.289589	14321.15

Table 3: Test data

9. LINGO CODE

9.1 LINGO CODE AND OUTPUT FOR 2030

```
MODEL:

! Objective function;
MIN = 6643.12*X_COAL
    + 14270.9*X_GAS
    + 6089.027*X_NUC
    + 23314.442*X_HYDRO
    + 14858.36*X_BIOM
    + 11907.042*SOL_TOTAL
    + 18407.224*WIND_TOTAL;

! Demand constraint;

  X_COAL + X_GAS + X_NUC + X_HYDRO + X_BIOM
  + SOL_TOTAL + WIND_TOTAL
  >= 5481.46*8760;

! Renewable share >= 50%;

  SOL_TOTAL + WIND_TOTAL + X_HYDRO + X_BIOM
  >= 2740.73*8760;

! Solar share >= 10%;

  SOL_TOTAL >= 548.146*8760;

! Wind share >= 5%;

  WIND_TOTAL >= 274.073*8760;

! Hydro share >= 15%;

  X_HYDRO >= 822.219*8760;

! Nuclear share >= 15%;

  X_NUC >= 822.219*8760;

! Coal share <= 35%;

  X_COAL <= 1918.511*8760;
X_COAL >= 1370*8760;
```

```
! Solar seasonality (fractions sum to 1);

  X_SOL_SUM = 0.35*SOL_TOTAL;


  X_SOL_MON = 0.15*SOL_TOTAL;


  X_SOL_AUT = 0.30*SOL_TOTAL;


  X_SOL_WIN = 0.20*SOL_TOTAL;

! Wind seasonality (fractions sum to 1);

  X_WIND_SUM = 0.15*WIND_TOTAL;
  X_WIND_MON = 0.45*WIND_TOTAL;
  X_WIND_AUT = 0.25*WIND_TOTAL;

  X_WIND_WIN = 0.15*WIND_TOTAL;

! Non-negativity;
X_COAL       >= 0;
X_GAS        >= 0;
X_NUC        >= 0;
X_HYDRO      >= 0;
X_BIOM       >= 0;
SOL_TOTAL    >= 0;
WIND_TOTAL   >= 0;
X_SOL_SUM    >= 0;
X_SOL_MON    >= 0;
X_SOL_AUT    >= 0;
X_SOL_WIN    >= 0;
X_WIND_SUM   >= 0;
X_WIND_MON   >= 0;
X_WIND_AUT   >= 0;
X_WIND_WIN   >= 0;

END
```

LINGO Result: OUTPUT FOR 2030

```
Licensee info: Eval Use Only
License expires: 14 APR 2026

Global optimal solution found.
Objective value:                              0.5364832E+12
Infeasibilities:                               0.000000
Total solver iterations:                              0
Elapsed runtime seconds:                           0.03

Model Class:                                         LP

Total variables:                     15
Nonlinear variables:                  0
Integer variables:                    0

Total constraints:                   32
Nonlinear constraints:                0

Total nonzeros:                      55
Nonlinear nonzeros:                   0
```

Variable	Value	Reduced Cost
X_COAL	0.1200120E+08	0.000000
X_GAS	0.000000	8181.873
X_NUC	0.1200759E+08	0.000000
X_HYDRO	7202638.	0.000000
X_BIOM	0.000000	2951.318
SOL_TOTAL	0.1440528E+08	0.000000
WIND_TOTAL	2400879.	0.000000
X_SOL_SUM	5041847.	0.000000
X_SOL_MON	2160792.	0.000000
X_SOL_AUT	4321583.	0.000000
X_SOL_WIN	2881055.	0.000000
X_WIND_SUM	360131.9	0.000000
X_WIND_MON	1080396.	0.000000
X_WIND_AUT	600219.9	0.000000
X_WIND_WIN	360131.9	0.000000

9.2 LINGO CODE AND OUTPUT FOR 2040

```
MODEL:

! Objective function;
MIN = 6643.12*X_COAL
    + 14270.9*X_GAS
    + 6089.027*X_NUC
    + 23314.442*X_HYDRO
    + 14858.36*X_BIOM
    + 11907.042*SOL_TOTAL
    + 18407.224*WIND_TOTAL;

! Demand constraint;

  X_COAL + X_GAS + X_NUC + X_HYDRO + X_BIOM
  + SOL_TOTAL + WIND_TOTAL
  >= 7930.331*8760;

! Renewable share >= 70%;

  SOL_TOTAL + WIND_TOTAL + X_HYDRO + X_BIOM
  >= 0.7*7930.331*8760;

! Solar share >= 25%;

  SOL_TOTAL >= 0.25*7930.331*8760;

! Wind share >= 12%;

  WIND_TOTAL >= 0.12*7930.331*8760;

! Hydro share >= 20%;

  X_HYDRO >= 0.2*7930.331*8760;

! Nuclear share >= 15%;

  X_NUC >= 0.15*7930.331*8760;

! Coal share <= 20%;

  X_COAL <= 0.2*7930.331*8760;
X_COAL >= 0.1*7930.331*8760;
```

```
! Solar seasonality (fractions sum to 1);

  X_SOL_SUM = 0.35*SOL_TOTAL;


  X_SOL_MON = 0.15*SOL_TOTAL;


  X_SOL_AUT = 0.30*SOL_TOTAL;


  X_SOL_WIN = 0.20*SOL_TOTAL;

! Wind seasonality (fractions sum to 1);

  X_WIND_SUM = 0.15*WIND_TOTAL;
  X_WIND_MON = 0.45*WIND_TOTAL;
  X_WIND_AUT = 0.25*WIND_TOTAL;

  X_WIND_WIN = 0.15*WIND_TOTAL;

! Non-negativity;
X_COAL       >= 0;
X_GAS        >= 0;
X_NUC        >= 0;
X_HYDRO      >= 0;
X_BIOM       >= 0;
SOL_TOTAL    >= 0;
WIND_TOTAL   >= 0;
X_SOL_SUM    >= 0;
X_SOL_MON    >= 0;
X_SOL_AUT    >= 0;
X_SOL_WIN    >= 0;
X_WIND_SUM   >= 0;
X_WIND_MON   >= 0;
X_WIND_AUT   >= 0;
X_WIND_WIN   >= 0;

END
```

LINGO Result: OUTPUT FOR 2040

```
LINGO/WIN64 21.0.38 (22 Apr 2025), LINDO API 15.0.6099.233

Licensee info: Eval Use Only
License expires: 14 APR 2026

Global optimal solution found.
Objective value:                              0.9224568E+12
Infeasibilities:                               0.000000
Total solver iterations:                              0
Elapsed runtime seconds:                           0.03

Model Class:                                         LP

Total variables:                     15
Nonlinear variables:                  0
Integer variables:                    0

Total constraints:                   32
Nonlinear constraints:                0

Total nonzeros:                      55
Nonlinear nonzeros:                   0


                      Variable           Value        Reduced Cost
                        X_COAL        6946970.            0.000000
                         X_GAS        0.000000            8181.873
                         X_NUC       0.1389394E+08        0.000000
                       X_HYDRO       0.1389394E+08        0.000000
                        X_BIOM        0.000000            2951.318
                     SOL_TOTAL       0.2639849E+08        0.000000
                    WIND_TOTAL        8336364.            0.000000
                     X_SOL_SUM        9239470.            0.000000
                     X_SOL_MON        3959773.            0.000000
                     X_SOL_AUT        7919546.            0.000000
                     X_SOL_WIN        5279697.            0.000000
                    X_WIND_SUM        1250455.            0.000000
                    X_WIND_MON        3751364.            0.000000
                    X_WIND_AUT        2084091.            0.000000
                    X_WIND_WIN        1250455.            0.000000
```

9.3 LINGO CODE AND OUTPUT FOR 2050

```
MODEL:

! Objective function;
MIN = 6643.12*X_COAL
    + 14270.9*X_GAS
    + 6089.027*X_NUC
    + 23314.442*X_HYDRO
    + 14858.36*X_BIOM
    + 11907.042*SOL_TOTAL
    + 18407.224*WIND_TOTAL;

! Demand constraint;

  X_COAL + X_GAS + X_NUC + X_HYDRO + X_BIOM
  + SOL_TOTAL + WIND_TOTAL
  >= 14321.15*8760;

! Renewable share >= 70%;

  SOL_TOTAL + WIND_TOTAL + X_HYDRO + X_BIOM
  >= 0.9*14321.15*8760;

! Solar share >= 40%;

  SOL_TOTAL >= 0.40*14321.15*8760;

! Wind share >= 15%;

  WIND_TOTAL >= 0.15*14321.15*8760;

! Hydro share >= 25%;

  X_HYDRO >= 0.25*14321.15*8760;

! Nuclear share >= 15%;

  X_NUC >= 0.15*14321.15*8760;

! Coal share <= 5%;

  X_COAL <= 0.05*14321.15*8760;
X_COAL >= 0.01*14321.15*8760;
```

```
! Solar seasonality (fractions sum to 1);

  X_SOL_SUM = 0.35*SOL_TOTAL;


  X_SOL_MON = 0.15*SOL_TOTAL;


  X_SOL_AUT = 0.30*SOL_TOTAL;


  X_SOL_WIN = 0.20*SOL_TOTAL;

! Wind seasonality (fractions sum to 1);

  X_WIND_SUM = 0.15*WIND_TOTAL;
  X_WIND_MON = 0.45*WIND_TOTAL;
  X_WIND_AUT = 0.25*WIND_TOTAL;

  X_WIND_WIN = 0.15*WIND_TOTAL;

! Non-negativity;
X_COAL       >= 0;
X_GAS        >= 0;
X_NUC        >= 0;
X_HYDRO      >= 0;
X_BIOM       >= 0;
SOL_TOTAL    >= 0;
WIND_TOTAL   >= 0;
X_SOL_SUM    >= 0;
X_SOL_MON    >= 0;
X_SOL_AUT    >= 0;
X_SOL_WIN    >= 0;
X_WIND_SUM   >= 0;
X_WIND_MON   >= 0;
X_WIND_AUT   >= 0;
X_WIND_WIN   >= 0;

END
```

LINGO Result: OUTPUT FOR 2050

```
LINGO/WIN64 21.0.38 (22 Apr 2025), LINDO API 15.0.6099.233

Licensee info: Eval Use Only
License expires: 14 APR 2026

Global optimal solution found.
Objective value:                              0.1947411E+13
Infeasibilities:                               0.000000
Total solver iterations:                              0
Elapsed runtime seconds:                           0.03

Model Class:                                         LP

Total variables:                     15
Nonlinear variables:                  0
Integer variables:                    0

Total constraints:                   32
Nonlinear constraints:                0

Total nonzeros:                      55
Nonlinear nonzeros:                   0


                      Variable           Value        Reduced Cost
                        X_COAL        1254533.            0.000000
                         X_GAS        0.000000            14270.90
                         X_NUC       0.1881799E+08        0.000000
                       X_HYDRO       0.3136332E+08        0.000000
                        X_BIOM        0.000000            2951.318
                     SOL_TOTAL       0.6272664E+08        0.000000
                    WIND_TOTAL       0.1881799E+08        0.000000
                     X_SOL_SUM       0.2195432E+08        0.000000
                     X_SOL_MON        9408996.            0.000000
                     X_SOL_AUT       0.1881799E+08        0.000000
                     X_SOL_WIN       0.1254533E+08        0.000000
                    X_WIND_SUM        2822699.            0.000000
                    X_WIND_MON        8468096.            0.000000
                    X_WIND_AUT        4704498.            0.000000
                    X_WIND_WIN        2822699.            0.000000
```

10. DISCUSSION AND RECOMMENDATIONS

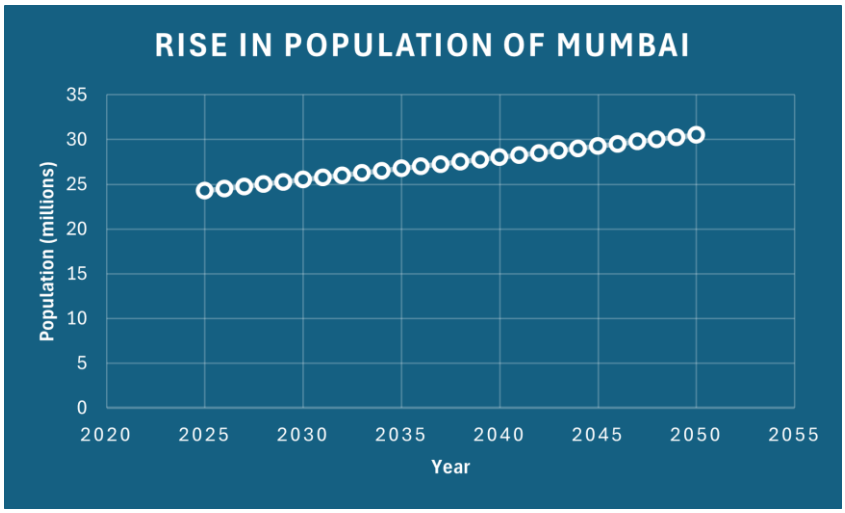

Figure 3: Trend showing a rise in the population of Mumbai

The above graph depicts the rise in Mumbai's population by 2050. It can be seen that the population is projected to **rise steadily from around 25 million in 2025 to 30 million by 2050.** It is already the 4th most populous city in the world and could rank 1st, according to predictions from several studies. The rise in population can stress Mumbai's resources, including power generation, and hence, renewable power capacity needs to be increased rapidly.

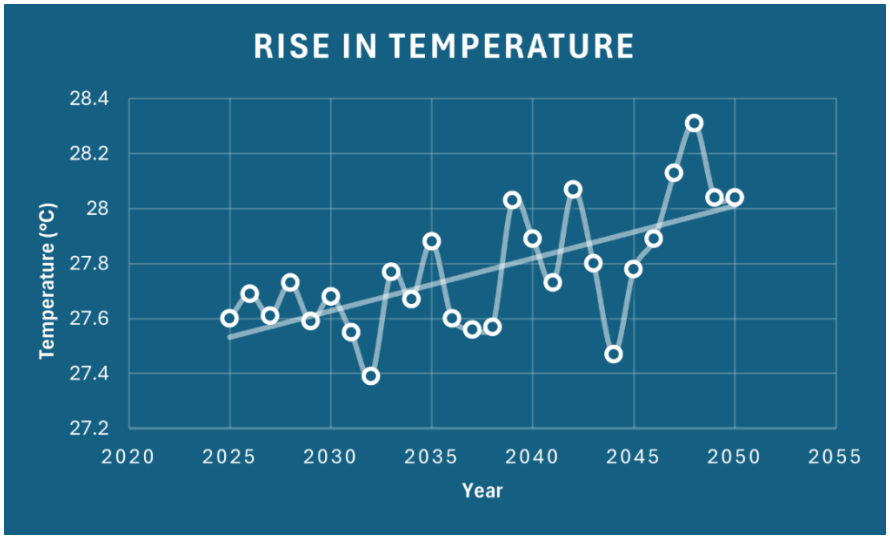

Figure 4: Trend showing a rise in the temperature of Mumbai

The above plot depicts the temperature values in the dataset considered for linear regression. The yearly temperature **variation is observed to be stochastic, but the general trend is one of increasing temperatures,** as expected. **It can be seen that the temperature is expected to**

rise by 1.25°C by 2050. This is a direct indication of global warming, mainly caused by the rising CO_2 levels in the atmosphere. Electricity generation is one of the biggest contributors to CO_2 rise due to dependence on coal power plants and hence needs to be replaced by low-emission power plants to keep CO_2 levels in check.

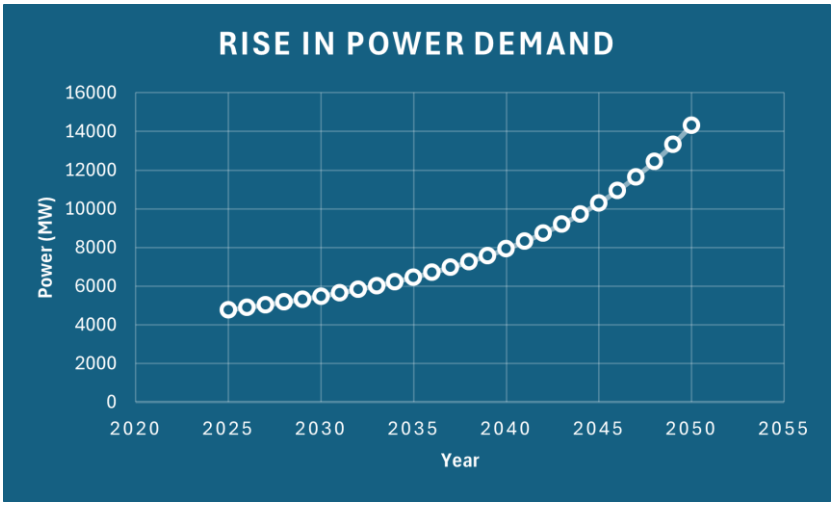

Figure 5: Trend showing a rise in power demand in Mumbai

Though the rise in population is linear, it can be observed from the above plot that the **power demand rise is non-linear.** Along with an increasing population, the lifestyle is also changing rapidly, due to which power demand is not proportional. Due to the rapid rise in electric vehicles, it can become one of the most power-consuming devices in the coming years. Along with electric vehicles, the electrification of several other products and processes contributes to the exponential increase in power demand. It reduces the consumption of legacy fuels such as gasoline and diesel due to the reduction in ICE vehicles, **but if the electricity is generated from coal and natural gas, electric vehicles become redundant.** Hence, it is essential to increase renewable energy capacities and phase out non-renewable power plants.

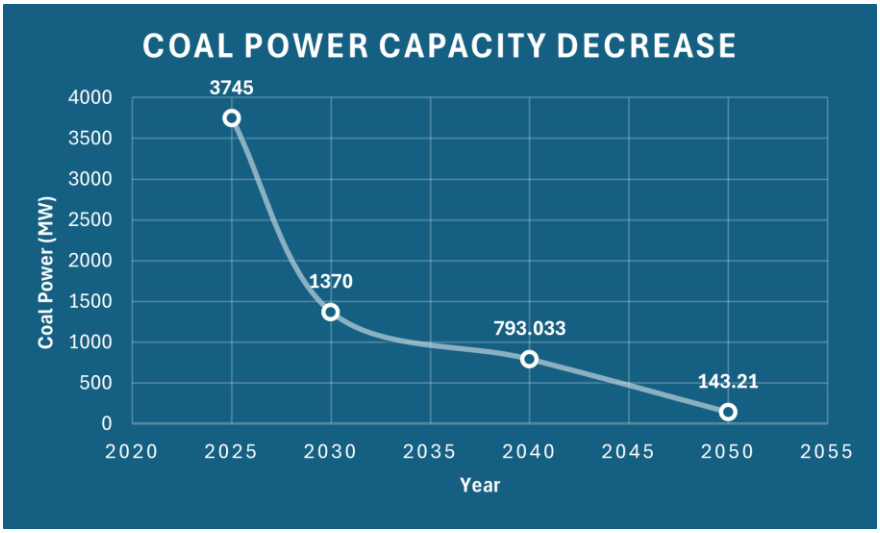

Figure 6: Required reduction in coal power

One of the main objectives of this research is to evaluate the amount of renewable energy capacity that needs to be added to cater to the rising electricity demand of Mumbai city, and hence reduce **the coal-fueled energy generation**. The reduction in coal power plants over the years is evident in the above graph. **Coal power is projected to decrease by close to 50% by 2030 and follow by a gradual decrease by 2050, reducing it to only 4% of its current capacity.** The 50% reduction in coal power generation by 2030 aligns with the 'Mumbai Climate Action Plan 2022' released by the government of Maharashtra.

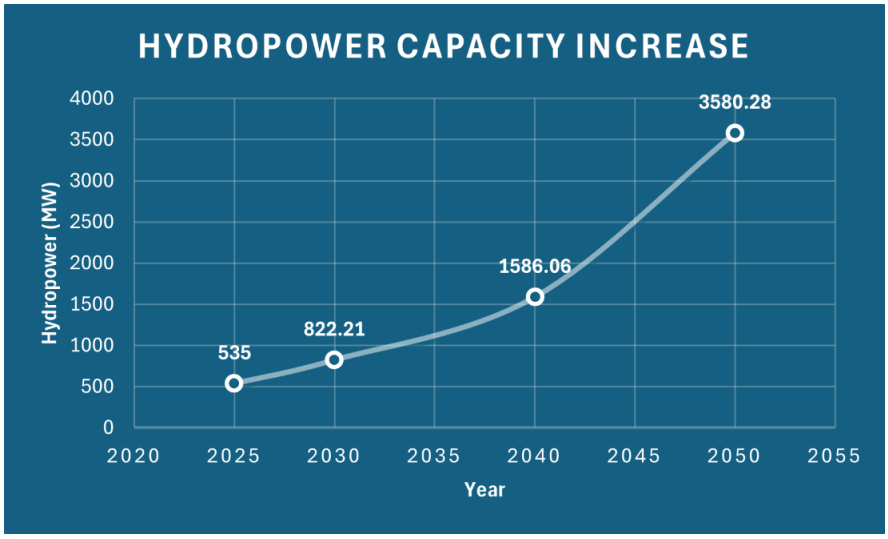

Figure 7: Required installation in Hydro power

Hydroelectric power plants have been leading the rise of renewable energy for more than a century, and they **remain one of the most feasible renewable power plants**. To achieve the target of generating 90% of the electricity from renewables by 2050, its capacity will need to rise steadily to cater to the rising demand for electricity. The Thansa River in the Thane district near Mumbai hold immense potential to generate hydroelectricity. **In fact, already 535 MW of power is being generated in this region, and more can be added to increase the capacity by 569% by 2050.**

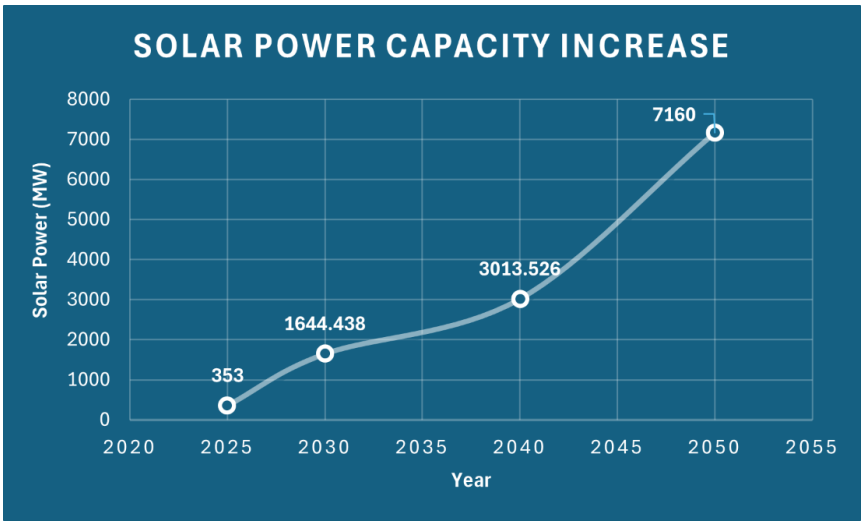

Figure 8: Required installation of Solar PV power

Solar power generation has been steadily rising in the last 25 years, and governments across the world have been striving to increase its capacity, including the Maharashtra government. This is possible due to a **reduction in cost and an increase in the overall efficiency of PV cells**. Solar power has not only proved to be a viable renewable energy source but also sustainable compared to other renewable energies. Hence, increasing its capacity is foundational to increasing the overall renewable energy power generation. **As per the 'Mumbai Climate Action Plan – 2022', solar power should contribute 40% of the total electricity demand; hence, the capacity is increasing to 7160 MW by 2050, as seen in the graph.** It needs to double every decade from 2030 to achieve this target. Most of this capacity can be added in the Raigarh district near Mumbai, which holds immense potential for solar power generation.

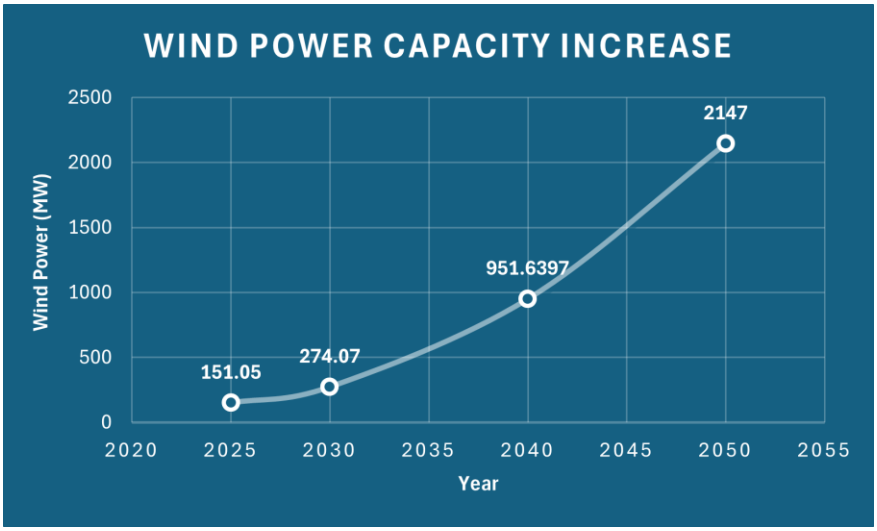

Figure 9: Required installation in Wind power

Wind power has also been one of the most important renewable powers and has risen steadily over the last century across the world, including India. **The wind turbine technology is expensive, which results in a lower economic feasibility** compared to other renewable power. Its **intermittency** in power generation also adds to its infeasibility. But over the last decade, wind turbine technology has been steadily improving and **becoming cheaper and efficient.** This has made **wind power more accessible,** and hence increasing its capacity is justified. From the above graph, it can be observed that wind power **capacity needs to increase exponentially** to meet the target of 15% of the total electricity generation. It needs to rise to 13 times the current capacity. Most of this capacity can again be added in Raigarh, which has lands at high altitude with high wind speeds.

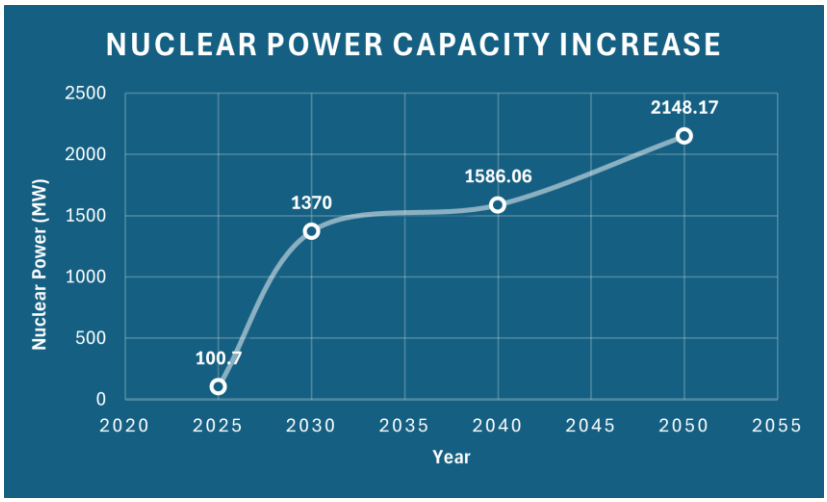

Figure 10: Required installation in nuclear power

Nuclear energy, though not considered renewable, has very low emissions compared to non-renewable energy sources and owing to the **immense Thorium reserves in India**, it is reasonable to increase its capacity. It can also generate large amounts of power and respond rapidly to varying loads. Mumbai is already home to the 'Bhabha Atomic Research Centre (BARC)', which was the first nuclear reactor in India. **Increasing its capacity will reduce Mumbai's dependence on coal and natural gas power plants. Due to the high thorium reserves, it allows India is not dependent on enriched uranium, making nuclear power cheaper in the long run.**

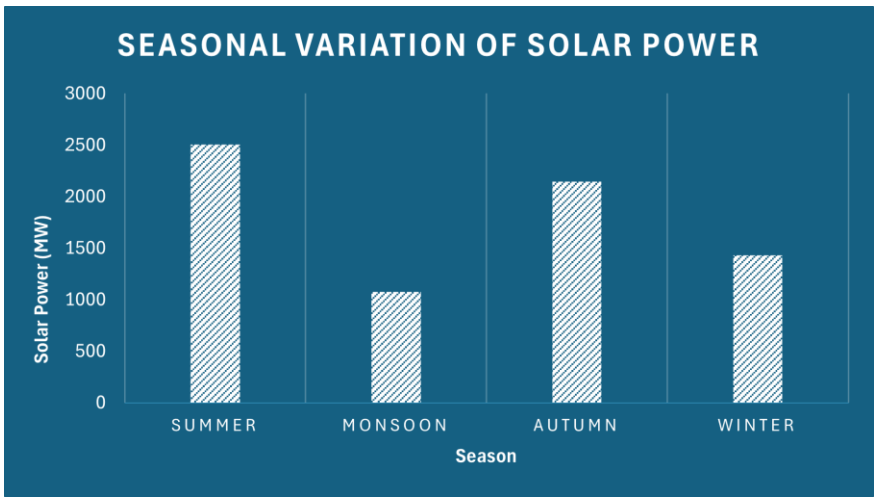

Figure 11: Seasonal variation of solar power

The above graph is a representation of seasonal variation in solar power output. This is due to the **varying insolation of sunlight on Earth throughout the year**. Summers have high insolation and hence higher power output. **Monsoons and winters can have cloud cover during the day, due to which the power output is comparatively lower compared to other seasons.** The image below shows the monthly variation in insolation in the northern and southern hemispheres.

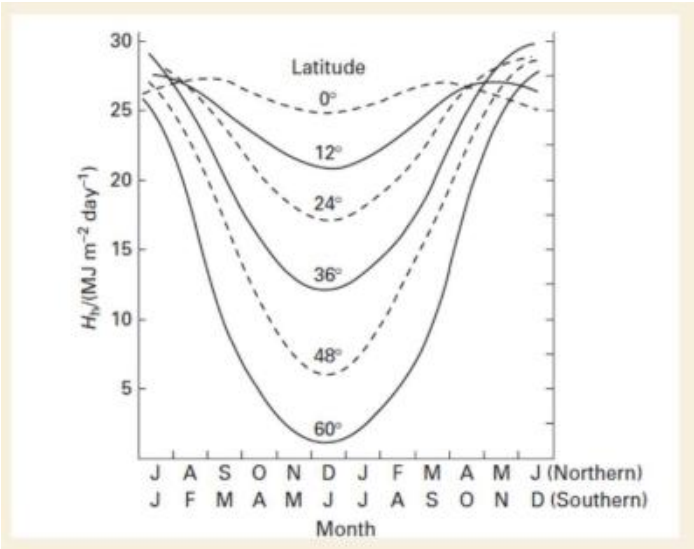

Figure 12: Variation in Insolation of sunlight across latitude (ME5480)

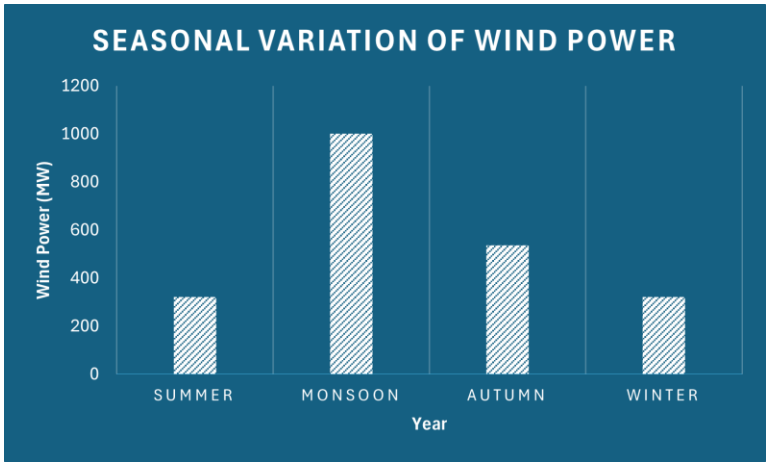

Figure 13: Seasonal variation of wind power

Wind power is also subjected to seasonal variations due to changes in the climate. Monsoons are accompanied by high wind speeds, where the average wind speed is higher than the rated wind speed of most turbines, so the power output is also high. **The high wind speeds are partially sustained through autumn, but rapidly drop during summer and winter seasons, which explains the low power output in these seasons.**

11. CONCLUSION

- **Environmental Sustainability:**
  - Use of natural gas has fallen a lot over the decades **because CO_2, NOx and SOx are so high.**
  - **Solar, wind, hydro and nuclear are the ideal mix**, given their tiny environmental footprints.
  - Model remains in compliance with MCAP renewable requirements (≥50% RE).
- **Economic Sustainability:**
  - The renewable resources (solar in particular) are optimal in the long run.
  - **High transport costs and emissions penalties** render coal and gas uneconomic.
  - Hydro and nuclear offer **lower emissions baseload** at stable long-term cost.
  - **Weighted objective (40 percent cost, 60 percent emissions) provides trade-offs between affordable and climate goals.**
- **Social Sustainability:**
  - **Decreased air pollution** results in **enhanced health and lower disease burden.**
  - Scaling renewable energy creates **high-sustainable jobs** in installation, O&M and grid upgrades.
  - **Resilience is enhanced** for all vulnerable populations when there is a cleaner energy system - low-income households can better withstand heat waves.
- **Learnings on Optimisation:**
  - **Trade-offs between cost**, **emission and policy constraints are** quantified in the optimisation.
  - Explicit formulation (variables, seasonal fractions, constraints, share) guarantees model stability.
  - **LP model provides transparency**: why some sources are chosen (hydro, nuclear) and some others discarded (gas, biomass).
- **Learnings on Forecasting:**
  - Forecasting demand using population, temperature and EV adoption ensures realistic input loads.
  - **Linking demand projections with supply modelling avoids under- or over-estimation of capacity needs.**

    - Seasonal patterns of solar/wind help allocate renewables more accurately.
- **Importance of Data & Formulation:**
    - **Consistent units (MWh),** reliable emission factors and correct **seasonal fractions** are critical.
    - A structured model prevents contradictions and ensures feasible solutions.
    - Accurate transport distances and LCOE values significantly influence cost-optimal outcomes.
- **Policy Alignment**
    - **Results align with MCAP 2022: high RE share, low coal, emphasis on storage-ready solar and wind.**
    - **Nuclear and hydro support a stable low-carbon baseload**, matching the national energy strategy.
    - **Coal cap (35%) and minimum shares (solar, wind, hydro, nuclear)** successfully drive clean transition.
- **Recommendations**
    - **Accelerate rooftop and utility-scale solar deployment.**
    - **Expand hydropower procurement and pumped storage** to stabilise RE-heavy grid.
    - **Strengthen nuclear capacity** as a long-term low-carbon baseload.
    - **Plan offshore wind** for post-2040 growth.
    - **Reduce coal reliance through phased shutdowns and cleaner alternatives.**
    - Improve grid infrastructure and transmission corridors for large-scale renewable integration.
    - Invest in long-duration storage and demand-side management.

12.REFERENCES

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